Do the app's motif labels name the tactic a position holds?

Three of the coaching app's board readers, scored on public puzzles

Chess Coach research lab

The main text records the earlier study at commit `d8b771238c`. Its results and examples are preserved. The appendix reports the later independent tests and a subsequent fixed combination at a different pin. The independent candidates were frozen before their held-out tests. The combination was fixed after those results and before its own scores.

Abstract

Three of the app's board readers put a motif word on a position: the theme of the best move the player missed (it files the misses deck, the puzzle buckets and the rule ledger, and the learner credits a solve to the skill it names), the motifs a bad move allowed (counted by the weakness profile, the metrics and the pattern census), and what a move does (the explanations' source). We scored each against the themes Lichess's own tagger put on 4,630 puzzles the app ships and on 20,000 held-out puzzles of the same ratings. All three name forks well. The misses deck's labeller has no word for pins, skewers, discovered attacks, deflections or trapped pieces, and when it says "defensive move" (26% of held-out puzzles) the tagger agrees 0.02 of the time. The review scan tags a hanging piece on 6,334 held-out puzzles, against 813 that the tagger says have one. The move describer is the most precise of the three and the one to build the others on: naming its skewers, discovered attacks and pins before the misses deck's defensive fallback raised agreement on both sets (8.3 and 5.6 points) and is kept.

1 The question

A motif word is a claim about the board. When the misses deck files a position under "defensive move", the learner counts a solve there toward a defensive-move skill; when the review scan tags a hanging piece, the weakness profile counts one more hanging piece. A wrong word sends practice and advice to the wrong place, so before the routines lean on these words we asked how often each is right when it is said (precision) and how often it is said when it is right (recall).

A Lichess puzzle is a mistake, its first move, followed by the line that punishes it, and Lichess tags each with themes. That is the shape the app's readers already read: the position the solver faces, the best move, the mistake that allowed it and the refutation. Lichess's tags come from another heuristic tagger, not from the truth, so this measures agreement; the three puzzles in Section 4 read disagreements on the board.

2 Method

Each reader runs on each puzzle the way the app runs it on a game (commit `d8b771238c`): the theme of the best move on the solver's position and the first solution move (`themeOfBest`); the

motifs of the mistake from the position before it, the mistake, the solution line and the score it reaches (`blunderMotifs`); and the findings that are a gain for the solver when the first solution move is described with the rest of the line as its reply (`describeMove`). Both sides' words are mapped to concepts by the app's own vocabulary, so a Lichess theme and an app word name one concept the way the app resolves them. A concept that only one side has a word for is not scored. The app's pool is the training set for any change tuned here; the held-out set, a seeded uniform sample of the same rating range that leaves out every puzzle the app ships, is the test.

3 Results

The misses deck's labeller. It has six motif words. When none fires it files the position under its phase (34% of the app's pool, 36% held out) or, when one of the solver's pieces was loose and the move protected it, as a defensive move. Puzzles are tactics, so the defensive move mostly catches the tactics the labeller has no word for: said on 864 of the app's pool and 5,166 held-out puzzles, it agrees with the tagger 0.03 and 0.02 of the time. Its hanging piece is right 0.60 and 0.28 of the time. It said back-rank mate on 0 of the app's pool and 12 held-out puzzles, 0 of them tagged so. Its fork is good.

The review scan. It tags a hanging piece whenever the refutation wins material, so it tags one on 1619 puzzles of the app's pool (609 tagged) and 6,334 held out (813 tagged): precision 0.38 and 0.13. It never tags a pin, a skewer, a back-rank mate or a trapped piece. Its fork and discovered attack are precise but catch about half.

The move describer. The most precise of the three on the tactics both vocabularies name: fork, skewer and discovered attack all above 0.6 precision and 0.5 recall on both sets. It finds a quarter of pins (0.24 held out), and its back-rank mate is mostly not one the tagger sees (0.18).

The vocabulary. The app's vocabulary does not resolve the named mates Lichess tags (`pillsburysMate`, `operaMate`, `epauletteMate`, `smotheredMate`, `cornerMate`, `swallowstailMate`, `morphysMate`, `hookMate`, `arabianMate`, `killBoxMate`, `triangleMate`, `anastasiaMate`, `blindSwineMate`, `bodenMate`, `dovetailMate`, `doubleBishopMate`, `vukovicMate`, `balestraMate`; 1,051 tags held out), so they score as nothing on either side.

	the app's pool				held out			
concept	tagged	said	precision	recall	tagged	said	precision	recall
checkmate	2074	1069	1.00	0.52	7833	3770	1.00	0.48
fork	563	389	0.92	0.63	2808	1804	0.91	0.58
discovered-attack	322	0	never said		1100	0	never said	
skewer	196	0	never said		511	0	never said	
pin	325	0	never said		1069	0	never said	
hanging-piece	609	682	0.60	0.67	813	2025	0.28	0.70
defensive-move	315	864	0.03	0.09	637	5166	0.02	0.14
back-rank-mate	346	0	never said		845	12	0.00	0.00
trapped-piece	170	0	never said		209	0	never said	
deflection	135	0	never said		789	0	never said	
attraction	113	0	never said		640	0	never said	
f7-raid	267	30	0.77	0.09	180	91	0.19	0.09
double-check	13	0	never said		80	0	never said	
zwischenzug	34	0	never said		214	0	never said	
removing-the-defender	5	0	never said		118	0	never said	
promotion	361	0	never said		468	0	never said	
passed-pawn	615	0	never said		1072	0	never said	

Table 1: The misses deck's labeller (`themeOfBest`). Tagged: puzzles Lichess tags with the concept; said: puzzles the reader names it on. Checkmate is said only for a mate in one.

	the app's pool				held out			
concept	tagged	said	precision	recall	tagged	said	precision	recall
fork	563	262	0.94	0.44	2808	1374	0.92	0.45
discovered-attack	322	178	0.90	0.50	1100	683	0.79	0.49
skewer	196	0	never said		511	0	never said	
pin	325	0	never said		1069	0	never said	
hanging-piece	609	1619	0.38	1.00	813	6334	0.13	1.00
defensive-move	315	0	never said		637	0	never said	
back-rank-mate	346	0	never said		845	0	never said	
trapped-piece	170	0	never said		209	0	never said	
deflection	135	61	0.46	0.21	789	548	0.55	0.38
attraction	113	146	0.18	0.24	640	883	0.23	0.32
f7-raid	267	0	never said		180	0	never said	
double-check	13	0	never said		80	0	never said	
zwischenzug	34	42	0.67	0.82	214	268	0.49	0.61
removing-the-defender	5	55	0.04	0.40	118	361	0.24	0.73
promotion	361	0	never said		468	0	never said	
passed-pawn	615	0	never said		1072	0	never said	

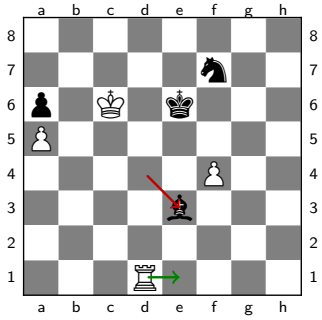
Table 2: The review scan's motifs (`blunderMotifs`). Checkmate is left out: the score passed in decides it.

	the app's pool				held out			
concept	tagged	said	precision	recall	tagged	said	precision	recall
checkmate	2074	0	never said		7833	0	never said	
fork	563	387	0.92	0.63	2808	1802	0.91	0.58
discovered-attack	322	246	0.79	0.61	1100	979	0.66	0.59
skewer	196	153	0.91	0.71	511	446	0.65	0.57
pin	325	111	0.73	0.25	1069	472	0.54	0.24
hanging-piece	609	0	never said		813	0	never said	
defensive-move	315	0	never said		637	0	never said	
back-rank-mate	346	263	0.21	0.16	845	774	0.18	0.17
trapped-piece	170	200	0.63	0.74	209	494	0.32	0.76
deflection	135	0	never said		789	0	never said	
attraction	113	0	never said		640	0	never said	
f7-raid	267	14	1.00	0.05	180	8	1.00	0.04
double-check	13	5	1.00	0.38	80	43	1.00	0.54
zwischenzug	34	0	never said		214	0	never said	
removing-the-defender	5	0	never said		118	0	never said	
promotion	361	16	1.00	0.04	468	34	0.97	0.07
passed-pawn	615	454	0.73	0.54	1072	806	0.64	0.48

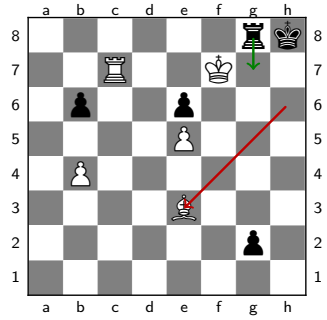
Table 3: The move describer's gain findings (`describeMove`).

4 Three held-out puzzles

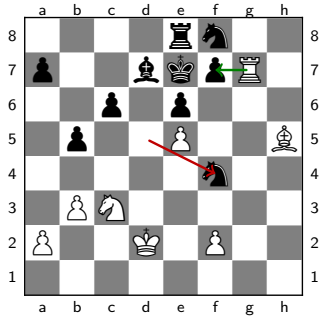
Each board is the position the solver faces; the red arrow is the mistake just played, the green one the first solution move.



Puzzle `0MHcM`, rated 1462



Puzzle `1dno2`, rated 1178



Puzzle `1Aofp`, rated 1168

0MHcM. After 42…♗e3, the solution 43. ♖e1 ♘e5+ 44. fxe5 (`d1e1 f7e5 f4e5`) from `8/5n2/p1K1k3/P7/5P2/4b3/8/3R4 w - - 3 43`. Lichess: `crushing`, `endgame`, `pin`, `short`. The misses deck's labeller: `defensiveMove`. The describer: `trapped-piece`, `pin`, `open-file`.

1dno2. After 41. ♗e3, the solution 41…♖g7+ 42. ♔f6 ♖xc7 (`g8g7 f7f6 g7c7`) from `6rk/2R2K2/1p2p3/4P3/1P6/4B3/6p1/8 b - - 1 41`. Lichess: `crushing`, `endgame`, `short`, `skewer`. The misses deck's labeller: `defensiveMove`. The describer: `skewer`.

1Aofp. After 34... ♘xf4, the solution 35. ♖xf7+ ♔d8 36. ♖xf4 (`g7f7 e7d8 f7f4`) from `4rn2/p2bkpR1/2p1p3/1p2P2B/5n2/1PN5/P2K1P2/8 w - - 0 35`. Lichess: `crushing`, `fork`, `middlegame`, `short`. The review scan: `hangingPiece`. The misses deck's labeller: `fork`.

5 A first candidate

The rule for a change, fixed before the first one ran: per puzzle, a reader *agrees* when one of its words names a concept the puzzle is tagged with and *disagrees* when one names a concept it is not tagged with; a change is kept when, on both sets, agreement rises with the 95% interval of the paired difference (2,000 bootstrap draws of puzzles) above zero and disagreement does not rise.

The first candidate changes only the misses deck's labeller: before its defensive fallback, it names a skewer, a discovered attack or a pin when the move describer finds one for the mover. Agreement rose from 40.7% to 49.0% on the app's pool (8.3 points, [7.5, 9.1]) and from 30.3% to 35.9% held out (5.6 points, [5.2, 5.9]); disagreement moved −2.0 points [−2.5, −1.5] and −0.2 points [−0.5, 0.1]. It is kept.

On the player's misses deck the same change re-files 44 positions, and the best line wins material or mates in 20 of them; the describer's pin, on 24 positions, wins material in 8, because it also fires on pins that win nothing at once. A port should carry the skewer and the discovered attack, and the pin only when the line wins with it.

6 What it means for the app

1. Port the kept candidate: the describer's skewer and discovered attack before the defensive fallback, and its pin only when the line wins with it; then make the defensive move require that the move answers a threat the board code can name, the next candidate.
2. The review scan's hanging piece should require a piece that was attacked and not defended before the refutation, not any material the refutation wins; its counts feed the weakness profile and the metrics.
3. The vocabulary should resolve the named mates, so a mate pattern is a concept on both sides.
4. Any change is tuned on the app's pool and kept only if it holds on the held-out set.

7 Limits

Lichess's tags are another tagger's, and a puzzle has as many tags as that tagger found, so a correct app word with no matching tag counts as wrong. Puzzles are tactics by construction; the misses deck is built from the player's own mistakes, where a defensive move is more common, which the drill-decks project measures on the deck itself.

A The later held-out test

This round measures app commit `f4bd63f19b343f686fb97ea4cf70b211c0bddc0a`. The app pool contains 4,630 unique shipped puzzles. The held-out sample contains 20,000 puzzles drawn uniformly from 4,682,966 eligible database rows, rated 400 through 1,899, excluding the app pool, with seed

20261014. The app-pool data hash is `005f31596fae53c51d76916faca5de4b02850d2beaea693e21e99ff574b40f85`; the serialized sample hash is `98c6777e6dbc7b81e0a7763d8005f3c12eb5ccc01da41a208103fa036614037a`, and the compressed database hash is `76335bfa7d7c4a7f93c1366d81549e53951ebb79dd43d34904cab8f22d962f8d`.

The independent candidates and measurement scripts were frozen at lab commit `da8a4cb1d3ef23ad4a71703dd09967`. Each candidate starts independently from the pinned app and changes its best-move labeller. The test uses the same frozen candidate sources and app-pool results. The generator checks those files against that commit before writing the paper. Those candidates were not retuned on the test.

The changes. The deflection candidate passes the supplied best line to the app's existing deflection reader. That reader requires a forcing move, the guard's departure and collection of the guarded target. It establishes a local capture-war gain, not the net value of a longer sacrifice. The trapped-piece candidate uses the move describer's gain finding, which requires that the piece loses where it stands and on every legal escape square. It follows that same piece through the supplied line until capture. The actual capture must win its capture war and leave a positive material gain after the preceding costs. Neither change searches every defense. Existing named tactics retain priority.

The unchanged rule. Agreement and disagreement are separate per-puzzle indicators: a said concept is present among the puzzle's tags, or a said concept is absent. The effect is the candidate-minus-pinned mean difference over the complete paired set. Keep only when, on both sets, the agreement difference's 95% interval is wholly positive and the disagreement difference's upper endpoint is at most 1 percentage point. The percentile intervals use 2,000 bootstrap draws of puzzles with replacement, each the original set size, with seed 20261014 restarted for each set. Both indicators use the same draws. The fixed rule has no multiple-comparison correction.

reader	set	agrees	disagrees
pinned	app pool	2,269	1,058
pinned	held out	7,179	6,759
deflection	app pool	2,308	1,044
deflection	held out	7,526	6,712
trapped piece	app pool	2,376	1,044
trapped piece	held out	7,324	6,769

Table 4: Puzzles on which the best-move labeller agrees or disagrees with a tag. Each reader scores the complete app pool and held-out sample.

candidate	set	agreement change	disagreement change	rule
deflection	app pool	0.84 [0.583, 1.102]	−0.30 [−0.562, −0.043]	pass
deflection	held out	1.74 [1.565, 1.920]	−0.24 [−0.420, −0.055]	pass
trapped piece	app pool	2.31 [1.901, 2.765]	−0.30 [−0.454, −0.151]	pass
trapped piece	held out	0.73 [0.615, 0.850]	0.05 [−0.020, 0.115]	pass

Table 5: Candidate-minus-pinned changes in percentage points, with 95% paired bootstrap intervals. Each candidate is tested independently against the pin. The last column applies the fixed rule to that set; keeping a candidate requires passing both sets.

The independent result. Both independent candidates meet the unchanged rule: `best-deflection`: keep; `best-trapped`: keep. The trapped-piece candidate's held-out disagreement point estimate rises. Its interval includes zero and its upper endpoint stays below the permitted limit. These separate results were observed before the combined test below.

The fixed combination. The combined candidate preserves both predicates and every existing tactic's priority. It tries `trappedPiece` before `deflection`, then the defensive and phase fallbacks. This is an explicit new tie-break: the pin has a move-describer gain stage but does not rank either new word. The combination extends that stage with the trapped-piece proof before trying the supplementary deflection reader. Its source, tests, priority, measurement scripts and prior results were bound by a content manifest before its scores. The manifest's hash is `01ee90236bcdf9928e70b827c3fdb9a8697373a4e395980791c6f64c3415c05e`. No alternative priority was scored.

The combined test uses the same app pool, held-out sample and unchanged keep rule. The individual held-out outcomes were already known. This is a test of the registered combination on those same sets, not a fresh independent sample. Its verdict is **keep**. The fixed combination passes on both sets and supports proposing this combined port.

set	agrees	disagrees
app pool	2,415	1,030
held out	7,671	6,722

Table 6: Combined best-move labeller: puzzles agreeing or disagreeing with a tag. The baseline counts appear in the independent results above.

set	agreement change	disagreement change	rule
app pool	3.15 [2.678, 3.650]	−0.60 [−0.907, −0.302]	pass
held out	2.46 [2.245, 2.680]	−0.18 [−0.385, 0.000]	pass

Table 7: Combined-minus-pinned changes in percentage points, with 95% paired bootstrap intervals, under the unchanged rule.

When both words are available. A word is available when its independent candidate emits it on that puzzle after all earlier tactics. Availability does not count raw findings that an earlier word hides. The overlap report checks every paired puzzle: use the first available new word in the registered order, or the pinned label when neither is available. The table counts the puzzles where both independent words are available and the combined word selected there. These counts are descriptive and add no decision threshold.

Neither measured set contains a puzzle where both independent candidates emit their new word, so there is no observed overlap winner. These data do not exercise the tie-break. The synthetic overlap regression covers that case.

What a port changes. The replacement misses publisher passes its existing SAN best line through the app's replay helper. A named miss can then carry `deflection` or `trappedPiece`; both

set	both available	trapped piece selected	deflection selected
app pool	0	0	0
held out	0	0	0

Table 8: Overlap of the independent new words and the combined selection. The registered order is trapped piece before deflection.

words already resolve to catalogue concepts. Neither has a direct `THEME_COMPETENCY` entry at the pin. Miss and graduation attempts use the existing `calc.scan` fallback, so these candidates do not add a deflection or trapped-piece skill credit. The separate routine facts caller still needs the optional line argument. Review motif code is unchanged, so the direct review weakness updates that consume those motifs are unchanged. A kept result supports proposing a port with its tests; it is not a record of a deployment.

Limits. The test measures agreement with Lichess's tagger on public puzzles. It does not measure changes in the player's misses deck, skill growth or results. The trapped-piece proof ends at the named capture, and the deflection proof is local to its collected target. Longer play and defenses outside the supplied line remain outside those claims.